

HW 5

① $a(t) = \cos(313\pi t) - \cos(113\pi t) - \cos(13\pi t)$ @ 100 Hz

highest frequency $\Rightarrow 313\pi$

$\omega_t > 2\omega_0$ therefore $\omega_{\text{sampling}} > 2 \cdot 313\pi \Rightarrow f_{\text{sampling}} > \frac{2 \cdot 313\pi}{2\pi}$
 $\uparrow$
 Sampling frequency
 $\Rightarrow > 313 \text{ Hz}$

② $a(t) = \cos(2\pi f t)$ @ 80 Hz $\Rightarrow a[n] = \cos(0.45\pi n)$

$\omega = 0.45\pi \Rightarrow 0.45\pi \cdot 80 = 2\pi f$

$\omega = \frac{2\pi f}{f_0}$

$f = 40 \cdot 0.45 \Rightarrow f_0 = 18 \text{ Hz}$
 $\uparrow$
 original frequency

③ $x[n] = \{2, -4, 2, -3\}$ $2 \leq n \leq 5$
 $y[n] = \{2, -2, -4, 2\}$ $-2 \leq n \leq 1$ $\rightarrow y[-n] = [2, -4, -2, 2]$ $-1 \leq n \leq 2$

$r_{xy}[n] = x[n] \otimes y[-n]$

n	-1	0	1	2	3	4	5
$x[n]$				2	-4	2	-3

$y[-n]$	2	-4	-2	2
(1)	4	-8	4	-6
(2)	-8	16	-8	12
	-4	8	-4	6
	4	-8	4	-6

$r_{xy}[n] = [4, -16, 16, -2, 0, 10, -6]$
 $1 \leq n \leq 7$

$\rho_{xy}[n] = \frac{r_{xy}[n]}{\sqrt{E_x E_y}}$ $E_x = 33$ $E_y = 28$

$\Rightarrow \frac{r_{xy}[n]}{30.397} \Rightarrow \rho_{xy}[n] = [0.13, -0.53, 0.53, -0.07, 0, 0.33, -0.20]$
 $1 \leq n \leq 7$